

Notes on Covariance Matrix Estimation and Linear Shrinkage [Lab3]

Politecnico di Milano
Financial Engineering, FY 2025-2026

1 Introduction

The covariance matrix Σ is ubiquitous in finance, but due to the properties of financial data its computation is non trivial and some points are worth discussing. There is a trade off between sample size, related to the estimate accuracy, and data non-stationarity:

- Financial data is non-stationary, i.e. the data generating distribution changes with time, meaning that more recent data is more important than past data. Thus, expanding estimates are not appropriate, since they incorporate different market structures/regimes;
- Volatility clusters (i.e. it is autocorrelated): rolling and exponential estimates are better suited to the task;
- The exponential estimate is more reactive, behaves more appropriately and is more theoretically founded.

When estimating the covariance matrix of a large universe of assets (few to tens of thousands) we have a large number of parameters and because of non-stationarity we will have few data points. The empirical covariance matrix S would be severely ill conditioned and non invertible, being dominated by noise and estimation errors.

Question 1. *Given an n dimensional investment universe, how many parameters are you estimating when computing the covariance matrix?*

2 A Solution: Linear Shrinkage

To address these issues, we introduce the concept of **shrinkage**. For simplicity we will focus on the sample covariance matrix. The intuition is simple: we impose some structure on the estimator to reduce estimation error. We do this by pulling the noisy sample covariance matrix towards a highly structured, biased estimator (the "target").

This is based on the bias-variance trade-off. The sample covariance matrix is unbiased but has high variance (estimation error). A structured target F is biased (misspecified) but has low variance (little estimation error). By combining them, we find an interior optimum that outperforms both. The idea is that by shrinking the extreme values of S (those most affected by noise) toward the structured estimator we get a better estimator.

The linear shrinkage estimator is defined as:

$$\hat{\Sigma}_{Shrink} = \delta F + (1 - \delta)S$$

where:

- S is the sample covariance matrix (unstructured, noisy).
- F is the shrinkage target (structured, stable).
- $\delta \in [0, 1]$ is the shrinkage intensity.

Geometrically, this can be interpreted as finding the optimal point along the segment connecting the sample matrix and the target matrix to minimize the distance to the true population matrix. The linear shrinkage methods differ in the choice of the structured estimator F and of the shrinkage intensity δ .

Question 2. *What happens qualitatively to the shrinkage intensity when using an exponential decay?*

3 Choosing the Target

The target should have a small number of free parameters (high structure) while reflecting important characteristics of asset returns. Two common targets proposed in the literature are:

3.1 The Single Factor Model

Proposed by Ledoit and Wolf (2002) [1], this target assumes stock returns are generated by a single factor (the market) through the model:

$$r_{it} = \alpha_i + \beta_i r_t^{mkt} + \epsilon_{it}$$

where residuals ϵ_{it} are uncorrelated to market returns r_t^{mkt} and to one another and $Var(\epsilon_{it}) = \sigma_i^2$ does not depend on time. The covariance matrix implied by this model is:

$$\Phi = \sigma_{mkt}^2 \beta \beta^T + \Delta$$

where σ_{mkt}^2 is the variance of market returns, $\beta = [\beta_i]_i$ and Δ is the diagonal matrix containing residual variances σ_i^2 .

- It reduces the estimation problem to estimating the market's variance, the asset betas and residual variances.
- It captures the dominant market factor but ignores extra-market covariance.

Question 3. *How can you estimate this model?*

3.2 The Constant Correlation Model

Proposed by Ledoit and Wolf (2003) [2], this target assumes all pairwise asset correlations are identical, leading to the model:

$$\Phi = [\phi_{ij}]_{ij}, \quad \phi_{ij} = \bar{\rho} \sqrt{\sigma_{ii} \sigma_{jj}} = \left(\frac{2}{n(n-1)} \sum_{i=1}^{n-1} \sum_{j=i+1}^n \rho_{ij} \right) \sqrt{\sigma_{ii} \sigma_{jj}},$$

where ρ_{ij} is the correlation between assets i and j .

- The correlation is estimated by the average of all sample correlations.
- It is extremely easy to implement and does not require identifying factors.
- The target matrix F has sample variances on the diagonal, but off-diagonal elements are derived from the average correlation.

Question 4. *How can you estimate this model?*

4 Determining the Shrinkage Intensity

The crucial step is determining the optimal weight δ . A value of $\delta = 0$ returns the noisy sample matrix; $\delta = 1$ returns the rigid target.

We seek the δ that minimizes the expected distance between our estimator and the true covariance matrix Σ . The distance is measured using the *Frobenius norm* (a quadratic loss function):

$$L(\delta) = \|\delta F + (1 - \delta)S - \Sigma\|^2$$

Analytical derivation shows that the optimal δ^* depends on three asymptotic constants:

1. $\pi = \sum_{i=1}^n \sum_{j=1}^n \text{AsyVar}(\sqrt{T}s_{ij})$ is the sum of asymptotic variances of the scaled entries of the sample covariance matrix (represents estimation error in S).
2. $\rho = \sum_{i=1}^n \sum_{j=1}^n \text{AsyCov}(\sqrt{T}f_{ij}, \sqrt{T}s_{ij})$ is the sum of asymptotic covariances between the entries of F and S .
3. $\gamma = \sum_{i=1}^n \sum_{j=1}^n (\phi_{ij} - \sigma_{ij})^2$ measures the misspecification (bias) of the shrinkage target.

The optimal shrinkage intensity is approximately:

$$\delta^* \approx \frac{\kappa}{T} = \frac{1}{T} \frac{\pi - \rho}{\gamma}$$

Importantly, we can estimate these constants consistently from the data itself. Notice that ρ depends on the shrinkage target:

1. $\hat{\pi} = \sum_{i=1}^n \sum_{j=1}^n \hat{\pi}_{ij}$, $\hat{\pi}_{ij} = \frac{1}{T} \sum_{t=1}^T [(r_{it} - \bar{r}_i)(r_{jt} - \bar{r}_j) - s_{ij}]$, $\bar{r}_* = \frac{1}{T} \sum_{t=1}^T r_{*t}$
2. $\hat{\gamma} = \sum_{i=1}^n \sum_{j=1}^n (f_{ij} - s_{ij})^2$
3. Single factor model:

$$\hat{\rho} = \sum_{i=1}^n \hat{\pi}_{ii} + \sum_{i=1}^n \sum_{j \neq i}^n p_{ij},$$

$$p_{ij} = \frac{1}{T} \sum_{t=1}^T \frac{s_{j,mkt} s_{mkt} (r_{it} - \bar{r}_i) + s_{i,mkt} s_{mkt} (r_{jt} - \bar{r}_j) - s_{i,mkt} s_{j,mkt} (r_{mkt,t} - \bar{r}_{mkt})}{s_{mkt}^2} \cdot (r_{mkt,t} - \bar{r}_{mkt}) (r_{it} - \bar{r}_i) (r_{jt} - \bar{r}_j) - f_{ij} s_{ij}$$

4. Constant correlation model:

$$\hat{\rho} = \sum_{i=1}^n \hat{\pi}_{ii} + \sum_{i=1}^n \sum_{j \neq i}^n \frac{\hat{\rho}}{2} \left(\sqrt{\frac{s_{jj}}{s_{ii}}} \hat{\vartheta}_{ii,ij} + \sqrt{\frac{s_{ii}}{s_{jj}}} \hat{\vartheta}_{jj,ij} \right),$$

$$\hat{\rho} = \frac{2}{n(n-1)} \sum_{i=1}^{n-1} \sum_{j=i+1}^n \frac{s_{ij}}{\sqrt{s_{ii} s_{jj}}}, \quad \hat{\vartheta}_{**,ij} = \frac{1}{T} \sum_{t=1}^T \{(r_{*t} - \bar{r}_*)^2 - s_{**}\} \{(r_{it} - \bar{r}_i)(r_{jt} - \bar{r}_j) - s_{ij}\}$$

Question 5. What happens to the shrinkage intensity when:

1. The estimation error is high?
2. The shrinkage target is badly misspecified?

5 Benefits in Practice

- **Invertibility:** Even if $n > T$ (where S is singular), the shrinkage estimator is positive definite and invertible, provided the target is positive definite.
- **Performance:** Empirical tests on historical data show that shrinkage estimators significantly reduce the out-of-sample variance of minimum-variance portfolios compared to multifactor models and the sample covariance matrix.

References

- [1] Ledoit, O., & Wolf, M. (2002). Improved estimation of the covariance matrix of stock returns with an application to portfolio selection *Journal of Empirical Finance*, 10 (2003), 603-621. URL: <http://www.ledoit.net/ole2.pdf>
- [2] Ledoit, O., & Wolf, M. (2003). Honey, I Shrunk the Sample Covariance Matrix. URL: <http://www.ledoit.net/honey.pdf>
- [3] Litterman, R., & Winkelmann, K. (1998). Estimating Covariance Matrices. *Goldman Sachs Risk Management Series*, January 1998. URL: https://people.duke.edu/~charvey/Teaching/BA453_2006/GS_Estimating_covariance_matrices.pdf